Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/16/2007
PI Name:	Lansford, Rusty Ph.D.		
Project Title:	Comparative Study Of Morphogenesis In 1g And Simulated Microgravity/Subproject A		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	91125	**Congressional District:**	26
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
Start Date:	01/01/2003	**End Date:**	12/31/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: per A. Chu, project is on NCE through 12/31/2006		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1610		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Microgravity modifies vital cell functions, including the cytoskeleton and cell shape, cell motility, gene expression, signal transduction and cell proliferation. These changes may cause health risk by reducing the efficiency of wound healing and immune response and increasing the frequency and invasiveness of cancer. The proposed interdisciplinary study of cell sorting will study some of these issues through the following hypotheses: 1) Altered gravity perturbs early developmental processes, such as cell sorting. 2) Cells sense gravity through their cytoskeleton. Altered gravity modifies cell motility and thus sorting through its effect on the cytoskeleton. 3) Microgravity effects on early development can be quantified by physical parameters such as viscoelasticity and surface tension, which relate to biologically relevant quantities such as strength of cell adhesion. Since studies in intact embryos are complicated, we will study aggregates of mixtures of chick embryo cells and cultured		

	cell lines (L and CHO) with known surface adhesion molecules. We will study cell sorting at 1g, in simulated microgravity using the Rotating Cell Culture System (RCCS), and in the presence of a drug (latrunculin) which disrupts the actin cytoskeleton. Confocal laser scanning and two-photon laser scanning microscopy will be used to track cells during sorting. Retroviruses will be employed to selectively label cells for tracking. Biomechanical techniques will be used to measure changes in relevant cell and tissue properties.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	A living body is more than a collection of randomly distributed cell types. Development involves not only the differentiation of cells, but also their organization into multicellular arrangements such as tissues and organs. Our studies address how cells interact with one another to form tissues and organs.
Task Progress:	My group is responsible for generating and providing desired fluorescent protein expressing retroviruses, for generating and isolating cadherin expressing cells by FACS, and for dynamically imaging the cell sorting experiments by multispectral two photon laser microscopy. All our goals were basically achieved by the end of year 3. Year 4 was spent primarily in support of Dr. Gabor's lab as they continued their experiments. Just the same, I have outlined some of the experimental goals of my proposal and summarized the progress made on them during year 3 of the NASA grant: Time-lapse videomicroscopy of fluorescently marked cells in cell populations Vital fluorescent marking of cells We have generated replication-defective, GFP expressing retroviruses to infect avian embryos. The HIV and Moloney based retroviruses have been pseudotyped with VSV-G (Zavada, 1982) in order to increase their host range and so they can be concentrated 1000-fold with minimal loss of biological activity (Burns et al., 1993). By uniting and modifying techniques and reagents previously reported (Kinsella and Nolan, 1996; Ory et al., 1996) we have obtained GFP expressing lentiviruses with titers ~ 1x109 infectious virions/ml. In year three, we placed far more emphasis in generating new HIV based lentiviruses that confer ubiquitous and tissue specific expression patterns of spectral distinct fluorescent proteins. We have used combinations of vital fluorescent dyes (DiI, DiO, and bodipy ceremide), fluorescent proteins (retrovirally expressed H2B-GFP, H2B-YFP, mito-YFP, and gap43-GFP), and fluorescent microbeads (0.04, 0.1, and 0.2 uM diameter beads, Molecular Probes) to fluorescently label L cells and CHO cells (data not shown). The L cells and CHO cells have been engineered to express various level of chick N-cadherin using chick N-cadherin retrovirus. Cells that express differing levels of N-cadherin on their cell surface have been isolated by FACS and by tissue culture limiting dilution methods. These cells have then been used to generate cellular spheroids for the cell sorting experiments. All of this work has been carried out in tight collaboration with Dr. Gabor Frogacs laboratory. Dynamic multispectral imaging of cell sorting We have designed a new generation of instrumentation, a multispectral detector (META) that allows the emission spectrum to be acquired from a single scan of the specimen (Patent approved November, 2001; "System and method for monitoring cellular activity"). The new multispectral detector has been engineered onto TPLSM that is also equipped with a movable stage, and is now commercially available as the Zeiss510 META. We have collected multispectral data in 3D from spheroid population of Ncadherin expressing cells that express multiple fluorophores. We are able to collect multispectral data simultaneously instead of sequentially using the 510 META, which decreases the light exposure (and bleaching) of the specimen and requires less time to collect an image set.
Bibliography Type:	Description: (Last Updated: 07/19/2007)
Articles in Peer-reviewed Journals	Cui C, Lansford R, Filla MB, Little CD, Cheuvront TJ, Rongish BJ. "Electroporation and EGFP labeling of gastrulating quail embryos." Dev Dyn. 2006 Oct;235(10):2802-10. PMID: 16894628 , Oct-2006